

Spectral Rigidity and Crystallization in the Coset Geometry of $\mathrm{PSL}(2,7)$

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Abstract

Let $G = \mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$ be the unique simple group of order 168. We study the coset geometry arising from the maximal subgroup $S_4 < G$ of index 7, and establish four main results. First, among the maximal subgroups of G , the subgroup S_4 — in either of its two conjugacy classes, the point- and line-stabilizers interchanged by the outer automorphism — is characterized by the property that the six-dimensional irreducible representation ρ_6 contains its trivial character exactly once (Theorem 2.3). Second, we provide explicit $\mathrm{GL}(3,2)$ generators for $S_4 = \mathrm{Stab}(P_1)$ in the Fano plane $\mathrm{PG}(2,2)$, with all presentation relations verified by direct matrix computation (Lemma 2.5). Third, the first non-trivial eigenvalue of the coset Laplacian on $L^2(G/S_4)$ is $\lambda_1 = 2/3$, and this spectral gap is stable under any G -invariant perturbation preserving the generating set, by virtue of the multiplicity-free decomposition $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$ and Schur's lemma (Lemma 2.9). Fourth, the random-walk Cheeger constant of the 7-vertex coset graph is $h = 8/21$, and the discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ holds, with neither bound attained (Theorem 3.3). As a separate structural result, we establish via Smith's theorem that $\mathrm{PSL}(2,7)$ cannot act freely on any sphere, so that $S^7/\mathrm{PSL}(2,7)$ is an orbifold, not a manifold. Explicit generators for the flag stabilizer $D_4 = \mathrm{Stab}(P_1, L_{124})$ are supplied in an appendix. The statements of Theorem 2.3 and Lemmas 2.8–2.9 are instances of classical facts: G acts 2-transitively on the seven cosets of S_4 , so (G, S_4) is a Gelfand pair, and the distance-transitive representations of the groups G with $\mathrm{PSL}_2(q) \triangleleft G \leq \mathrm{P}\Gamma\mathrm{L}_2(q)$ are classified [15]; this note records the case $q = 7$ in spectral language.

These results are of independent mathematical interest; their potential physical interpretation requires additional theoretical development and is not assumed within the scope of this work.

Keywords: $\mathrm{PSL}(2,7)$; finite simple groups; Fano plane; coset geometry; spectral gap; Cheeger constant

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1. Introduction

The group $G = \mathrm{PSL}(2,7)$, the projective special linear group over the field of seven elements, is the unique simple group of order $168 = 2^3 \cdot 3 \cdot 7$. Through the isomorphism $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$ [4], it also acts faithfully as the automorphism group of the Fano plane $\mathrm{PG}(2, \mathbb{Z}_2)$, the projective plane over $\mathbb{Z}_2$ whose seven points and seven lines admit a transitive, indeed 2-transitive, collineation group of order 168. This dual identity — as a group of Möbius transformations on $\mathbb{P}^1(\mathbb{Z}_7)$ and as a group of projective symmetries of the smallest projective plane — makes $\mathrm{PSL}(2,7)$ a natural meeting point of classical number theory, finite combinatorics, and spectral geometry.

The maximal subgroup structure of G is completely determined by Dickson's classification of the maximal subgroups of $\mathrm{PSL}(2,q)$ for prime powers q [6, 7]. For $q = 7$, there are exactly three conjugacy classes of maximal subgroups, comprising two classes of S_4 and one of the Borel subgroup:

$$S_4 = \mathrm{Stab}(P) \text{ (point-stabilizer, order 24, index 7), } S_4 = \mathrm{Stab}(\ell) \text{ (line-stabilizer, order 24, index 7), } \\ Z_7 \rtimes Z_3 \text{ (order 21, index 8).}$$

Among these, the symmetric group S_4 occupies a distinguished spectral position that we call *spectral crystallization*: among the maximal subgroups of G , the property that the 6-dimensional irreducible representation ρ_6 of G contains the trivial representation of H with multiplicity exactly one picks out precisely the two conjugacy classes of S_4 , the point- and line-stabilizers in $\mathrm{PG}(2,2)$, and excludes the Borel subgroup $Z_7 \rtimes Z_3$.

The first goal of this paper is to establish this characterization precisely, by direct Frobenius reciprocity calculations for each maximal subgroup class (Theorem 2.3), and to record that the two S_4 classes are interchanged by the outer automorphism of G — the algebraic form of Fano-plane point-line duality. The second goal is to realize the characterization concretely within the Fano plane geometry, providing explicit $\mathrm{GL}(3,2)$ generator matrices for S_4 as the point stabilizer $\mathrm{Stab}(P_1)$, with all presentation and order relations verified by direct matrix computation over $\mathbb{Z}_2$ (Lemma 2.5).

The spectral content of the coset space G/S_4 is simple: by Frobenius reciprocity, $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$, with all other irreducible representations of G absent. The 7-dimensional coset Laplacian therefore has exactly one non-trivial eigenvalue $\lambda_1 = 2/3$, computed from the character value $\chi_{\rho_6}(2A) = 2$ via Schur's lemma. Lemma 2.9 shows this gap is rigid: any G -invariant perturbation of the Laplacian preserving the generating set cannot alter the eigenvalue on the ρ_6 -isotypic component.

In Section 3 we analyze the 7-vertex coset graph. The 2-transitive action of G on $\mathrm{PG}(2,2)$ forces the graph to be the complete multigraph $2 \cdot K_7$ (every pair of vertices connected by exactly two edges). We compute the random-walk Cheeger constant $h = 8/21$ and verify the discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ [3], noting that neither bound is attained.

In Section 4 we maintain a careful separation among three spectral quantities that arise naturally in this geometry: the coset Laplacian eigenvalue $\lambda_1 = 2/3$, the inter-representation gap $\delta = 10/21$ in the full $\mathrm{PSL}(2,7)$ spectrum, and the Cheeger constant $h = 8/21$. These are conceptually distinct objects, and we flag the numerical coincidence $h_2 = \delta = 10/21$ (where h_2 denotes the Cheeger ratio at 2-

vertex cuts) as a structural open question. We also establish, as an independent result via Smith's theorem, that $\mathrm{PSL}(2,7)$ cannot act freely on any sphere, so $S^7/\mathrm{PSL}(2,7)$ is an orbifold. The orbifold index theory of this space is identified as an open problem requiring methods beyond the scope of this paper.

Prior art. The spectral statements of Section 2 are instances of classical facts, recorded here for $q = 7$. The group G acts 2-transitively on the seven cosets of S_4 , so the permutation character is $1 + \chi_{\rho_6}$ (Lemma 2.8) and (G, S_4) is a Gelfand pair; the rigidity of Lemma 2.9 is Schur's lemma for this rank-2 action, and the eigenvalue $2/3$ is read off from the central character of the class $2A$ on ρ_6 . Multiplicity-free permutation representations of the finite linear groups were studied by Inglis, Liebeck and Saxl [16]. Praeger, Saxl and Yokoyama [18] reduced the classification of primitive distance-transitive graphs to the almost simple and affine cases, and Faradžev and Ivanov [15] classified the distance-transitive representations of the groups G with $\mathrm{PSL}_2(q) \triangleleft G \leq \mathrm{P}\Gamma\mathrm{L}_2(q)$. The terms *spectral rigidity* and *spectral crystallization* used in this note are descriptive names for these facts; no further theorem is attached to them, and no novelty is claimed for these statements.

Conventions. All groups are finite. Matrices are written over $\mathbb{Z}_2$ unless stated otherwise, with arithmetic modulo 2. The inner product of class functions χ, ψ on a finite group H is $\langle \chi, \psi \rangle_H = (1/|H|) \sum_{h \in H} \chi(h)\psi(h)$. Conjugacy class labels ($1A$, $2A$, etc.) follow the ATLAS of Finite Groups [5]. The character table of $\mathrm{PSL}(2,7)$ appears in [5] and is recalled in Table 1 below.

2. Main Results

2.1 Subgroup Classification

We recall the relevant structure of $G = \mathrm{PSL}(2,7)$.

Theorem 2.1 (Dickson [6]; Huppert [7], Satz II.8.27; Wilson [12], §3.3.5). *The maximal subgroups of $\mathrm{PSL}(2,7)$, up to conjugacy, are exactly:*

Label	Isomorphism type	Order	Index
M_1	$S_4 = \mathrm{Stab}(P)$	24	7
M_2	$Z_7 \rtimes Z_3$	21	8
M_3	$S_4 = \mathrm{Stab}(\ell)$	24	7

Remark 2.2. The two classes of S_4 are the point-stabilizers $\mathrm{Stab}(P)$ and the line-stabilizers $\mathrm{Stab}(\ell)$ in $\mathrm{PG}(2,2)$. They are abstractly isomorphic but not conjugate in G : the outer automorphism of $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$, realized algebraically as the inverse-transpose map $g \mapsto (g^{-1})^T$ — the Fano-plane point-line duality — interchanges them, while fixing the Borel class $Z_7 \rtimes Z_3$ (the Singer-cycle normalizer). The dihedral group D_4 of order 8 is the Sylow 2-subgroup of G and is not maximal: since $7 \equiv -1 \pmod{8}$, G contains S_4 , and D_4 is the Sylow 2-subgroup of each S_4 , so $D_4 \subsetneq S_4 \subsetneq G$. It appears in this work only as the flag stabilizer $\mathrm{Stab}(P_1, L_{124})$ of index 21 (Appendix A), not as a maximal subgroup.

The irreducible complex representations of G have dimensions 1, 3, 3, 6, 7, 8, which we denote ρ_1 , ρ_3 , ρ_3' , ρ_6 , ρ_7 , ρ_8 respectively. The dimension identity $1 + 9 + 9 + 36 + 49 + 64 = 168$ confirms six conjugacy classes.

Table 1: Conjugacy class sizes and ρ_6 character values for $PSL(2,7)$ [5].

Class	1A	2A	3A	4A	7A/7B
Class size	1	21	56	42	24 each
χ_{ρ_6}	6	2	0	0	-1

Remark 2.2b (Fusion of conjugacy classes). For any prime $p \geq 5$, the group $PSL(2,p)$ has a unique conjugacy class of involutions. In $G = PSL(2,7)$, there is therefore a unique class of elements of order 2, labeled 2A, of size 21. Consequently, every element of order 2 in any subgroup $H < G$ fuses to class 2A under the inclusion $H \hookrightarrow G$ [1]. Similarly, G has a unique class 3A of elements of order 3 and a unique class 4A of elements of order 4, so elements of those orders in subgroups fuse accordingly.

2.2 The S_4 Characterization Theorem

For a subgroup $H < G$ and an irreducible representation (ρ, V) of G , define the *trivial multiplicity*

$$n_1(H; \rho) = \langle \chi_\rho|_H, 1_H \rangle_H = (1/|H|) \sum_{h \in H} \chi_\rho(h).$$

By Frobenius reciprocity [10, Theorem 7.2], $n_1(H; \rho)$ equals the multiplicity of ρ in $\text{Ind}^H 1_H \cong L^2(G/H)$.

Theorem 2.3 (S_4 Characterization). *Among the maximal subgroups of $G = PSL(2,7)$, the subgroup S_4 — in each of its two conjugacy classes (the point- and line-stabilizers, interchanged by the outer automorphism) — satisfies $n_1(H; \rho_6) = 1$, and these are the only maximal subgroups with this property; the Borel subgroup $Z_7 \rtimes Z_3$ has $n_1(H; \rho_6) = 0$. Moreover, the two classes of S_4 are exactly the maximal subgroups of G whose order is divisible by 3 and which contain elements of order 4.*

Proof. We check the maximal subgroup classes from Theorem 2.1, using the character values in Table 1 and the fusion of Remark 2.2b. Since $n_1(H; \rho)$ depends only on the conjugacy classes of G met by H (through the character χ_ρ), it is constant across isomorphic subgroups with identical fusion; in particular the two S_4 classes share the same value.

Case 1: $H = M_1 = S_4$ (point stabilizer), $|H| = 24$. The group S_4 has element orders: 1 identity, 6 transpositions, 3 double transpositions (2-cycles paired), 8 elements of order 3, and 6 elements of order 4. All 9 elements of order 2 fuse to 2A; the 8 elements of order 3 fuse to 3A; the 6 elements of order 4 fuse to 4A. Therefore:

$$n_1(S_4; \rho_6) = (1/24)[1 \cdot 6 + 9 \cdot 2 + 8 \cdot 0 + 6 \cdot 0] = 24/24 = 1. \quad \checkmark$$

Case 2: $H = M_2 = Z_7 \rtimes Z_3$, $|H| = 21$. The group $Z_7 \rtimes Z_3$ has element orders: 1 identity, 14 elements of order 3 (fusing to 3A), and 6 elements of order 7 (fusing to class 7A or 7B). There are no elements of order 2 or 4. Therefore:

$$n_1(M_2; \rho_6) = (1/21)[1 \cdot 6 + 14 \cdot 0 + 6 \cdot (-1)] = 0/21 = 0.$$

Case 3: $H = M_3 = S_4$ (line stabilizer), $|H| = 24$. The line-stabilizer is abstractly isomorphic to the point-stabilizer of Case 1, with the same element-order distribution and the same fusion into the classes 2A, 3A, 4A of G . The outer automorphism $g \mapsto (g^{-1})^T$ carries M_1 to M_3 and fixes the character χ_{ρ_6} , so:

$$n_1(M_3; \rho_6) = n_1(M_1; \rho_6) = 1. \quad \checkmark$$

Thus $n_1(H; \rho_6) = 1$ for both S_4 classes (M_1 and M_3), while $n_1(M_2; \rho_6) = 0$ for the Borel subgroup. Among the maximal subgroups of G , the value $n_1(\cdot; \rho_6) = 1$ therefore characterizes exactly the S_4 isomorphism type. The divisibility statement follows similarly: both S_4 classes have order $24 = 2^3 \cdot 3$ and contain elements of order 4, whereas the Borel M_2 has order $21 = 3 \cdot 7$ and no element of order 4.

□

Remark 2.3b. The condition $n_1(\cdot; \rho_6) = 1$ is genuinely restrictive even among small subgroups. The (non-maximal) Sylow 2-subgroup $D_4 < S_4$, which appears as the flag stabilizer of Appendix A, has 1 identity, 5 involutions (fusing to 2A), and 2 elements of order 4 (fusing to 4A), giving $n_1(D_4; \rho_6) = (1/8)[1 \cdot 6 + 5 \cdot 2 + 2 \cdot 0] = 16/8 = 2$. Thus the value 1 is not automatic for subgroups that contain both involutions and order-4 elements; it is specific to the maximal S_4 classes.

2.3 Constructive $GL(3,2)$ Generators for S_4

We realize the S_4 characterization concretely via the Fano plane geometry.

Definition 2.4 (Fano plane). The Fano plane $PG(2,2)$ consists of the 7 nonzero vectors in $\mathbb{Z}_2^3$, labeled:

$$P_1 = (1,0,0), P_2 = (0,1,0), P_3 = (0,0,1), P_4 = (1,1,0), P_5 = (1,0,1), P_6 = (0,1,1), P_7 = (1,1,1).$$

Its 7 lines are the triples of points whose (mod-2) sum is zero. The four lines of $PG(2,2)$ not passing through P_1 are:

$$L_1 = \{P_2, P_3, P_6\}, L_2 = \{P_2, P_5, P_7\}, L_3 = \{P_3, P_4, P_7\}, L_4 = \{P_4, P_5, P_6\}.$$

Lemma 2.5 (Constructive S_4 embedding). The point stabilizer $Stab_{GL(3,2)}(P_1)$ is isomorphic to S_4 and is generated over $\mathbb{Z}_2$ by the matrices

$$g_1 = [[1,0,0],[0,0,1],[0,1,0]], \quad g_2 = [[1,1,0],[0,1,1],[0,0,1]]$$

satisfying the presentation $\langle g_1, g_2 \mid g_1^2 = g_2^4 = (g_1 g_2)^3 = I \rangle \cong S_4$.

Proof. We verify each claim by direct computation over $\mathbb{Z}_2$.

(1) *Membership in $GL(3,2)$.* $\det(g_1) = 1(0 \cdot 0 - 1 \cdot 1) = 1 \pmod{2}$ and $\det(g_2) = 1(1 \cdot 1 - 1 \cdot 0) = 1$. Both matrices are invertible over $\mathbb{Z}_2$.

(2) *Stabilization of $P_1 = (1,0,0)^T$.* $g_1(1,0,0)^T = (1,0,0)^T = P_1$ and $g_2(1,0,0)^T = (1,0,0)^T = P_1$.

(3) *Orders.* $g_1^2 = I$ (verified by direct multiplication; g_1 is a reflection). $g_2^2 = [[1,0,1],[0,1,0],[0,0,1]] \neq I$, but $g_2^4 = I$, confirming g_2 has order 4.

(4) *Mixed relation.* One computes $g_1 g_2 = [[1,1,0],[0,0,1],[0,1,1]]$, and verifies $(g_1 g_2)^3 = I$ by direct matrix multiplication.

(5) *Faithful action on $\{L_1, L_2, L_3, L_4\}$.* Applying g_1 and g_2 to the four lines: $g_1 \mapsto (L_2 L_3)$, $g_2 \mapsto (L_1 L_4 L_2 L_3)$. These permutations have orders 2 and 4 respectively. Their product $(L_2 L_3)(L_1 L_4 L_2 L_3) = (L_1 L_4 L_3)$ has order 3, satisfying the presentation. The group $\langle g_1, g_2 \rangle$ therefore surjects onto S_4 . Since $|\text{Stab}(P_1)| = |\text{GL}(3,2)|/7 = 168/7 = 24 = |S_4|$, the surjection is an isomorphism.

□

2.4 Spectral Decomposition and Gap Stability

The coset space G/S_4 has 7 points, identified with the points of the Fano plane $PG(2,2)$.

Definition 2.6 (Coset Laplacian). Let $S = \{s \in G : s \text{ lies in class } 2A\}$, so $|S| = 21$. The coset Laplacian Δ is the operator on $L^2(G/S_4)$ defined by

$$(\Delta f)(P) = f(P) - (1/|S|) \sum_{s \in S} f(s \cdot P)$$

for $f : G/S_4 \rightarrow \mathbb{C}$ and $P \in G/S_4$.

Remark 2.7. Since G acts 2-transitively on the 7 points of $PG(2,2)$, each of the 21 class-2A involutions either fixes a coset or transposes two cosets. For a fixed coset P , the 9 involutions in S_4 fix P , and the remaining 12 move P to one of 6 other cosets (by 2-transitivity, exactly 2 involutions send P to each specific other coset Q). Thus $\sum_{s \in S} f(s \cdot P) = 9f(P) + 2 \sum_{Q \neq P} f(Q)$, and for $f \perp 1$:

$$(\Delta f)(P) = f(P) - (9f(P) - 2f(P))/21 = f(P) - 7f(P)/21 = (2/3)f(P).$$

Lemma 2.8 (Spectral decomposition). $L^2(PSL(2,7)/S_4) \cong \rho_1 \oplus \rho_6$. All other irreducible representations of $PSL(2,7)$ (of dimensions 3, 3, 7, 8) occur with multiplicity zero.

Proof. By Frobenius reciprocity [10, Theorem 7.2], the multiplicity of an irreducible ρ in $L^2(G/S_4) \cong \text{Ind}(S_4, G) \otimes 1$ equals $n_1(S_4; \rho)$. From the proof of Theorem 2.3: $n_1(S_4; \rho_6) = 1$. The trivial representation gives $n_1(S_4; \rho_1) = 1$ trivially. For the remaining four irreps, direct inner product calculations yield $n_1(S_4; \rho_3) = n_1(S_4; \rho_3') = n_1(S_4; \rho_7) = n_1(S_4; \rho_8) = 0$. The dimension check $1 + 6 = 7 = |G/S_4|$ confirms completeness.

□

Lemma 2.9 (Spectral gap and stability). The first non-trivial eigenvalue of Δ is $\lambda_1 = 2/3$. Moreover, every G -invariant operator on $L^2(G/S_4)$ acts as a scalar on the ρ_6 -isotypic component; the eigenvalue $2/3$ is an exact, not approximate, spectral identity for the specific operator Δ .

Proof. Step 1 (Schur's lemma). By Lemma 2.8, $L^2(G/S_4) \cong \rho_1 \oplus \rho_6$ with each irreducible appearing with multiplicity one. By Schur's lemma [10, Theorem 1.7], $\text{End}^G(L^2(G/S_4)) \cong \mathbb{C} \oplus \mathbb{C}$. Every G -equivariant operator on $L^2(G/S_4)$ acts as a scalar on each isotypic component.

Step 2 (G-invariance of Δ). The generating set S consists entirely of conjugacy class $2A$, so S is closed under conjugation by G . Therefore Δ commutes with the left G -action: Δ is G -invariant. Any G -invariant perturbation $\Delta + V$ (where V commutes with the G -action) likewise decomposes into scalars on ρ_1 and ρ_6 .

Step 3 (Eigenvalue computation). All 21 elements of S lie in class $2A$. By Schur's lemma, the operator $A = (1/|S|)\sum_{s \in S} \rho_6(s)$ equals a scalar $c \cdot I_6$ on ρ_6 . Taking traces: $c = \chi_{\rho_6}(2A)/\dim(\rho_6) = 2/6 = 1/3$. Therefore $\Delta = I - A$ acts on ρ_6 as $\lambda_1 = 1 - c = 2/3$.

Step 4 (Stability statement). The decomposition $\rho_1 \oplus \rho_6$ is determined by group theory alone and cannot be altered by G -invariant perturbations. A different G -invariant operator (with a different generating set) would preserve the decomposition but alter the scalar c ; the value $\lambda_1 = 2/3$ is the exact eigenvalue for the specific operator Δ defined above.

□

3. Cheeger Analysis

3.1 Structure of the Coset Graph

Definition 3.1 (Coset graph). The coset graph Γ has vertex set $V = G/S_4$, the 7 cosets of S_4 , identified with $\{P_1, \dots, P_7\}$. For each class- $2A$ involution s and each $P \in V$, place a directed edge $(P, s \cdot P)$.

Proposition 3.2. *The underlying undirected coset graph Γ is the complete multigraph $2 \cdot K_7$: for every pair $\{P, Q\}$ with $P \neq Q$, the vertices P and Q are connected by exactly 2 edges. Each vertex has degree 12.*

Proof. Since G acts 2-transitively on the 7 points of $PG(2,2)$, the number of involutions sending a fixed P to a fixed $Q \neq P$ is constant over all ordered pairs (P, Q) . The total number of involution-vertex pairs (s, P) with $s \cdot P \neq P$ equals $|S| \cdot 7 - (\text{fixed pairs})$. The 21 involutions fix coset P_j if and only if $s \in S \cap S_4^{g_j}$ (the conjugate of S_4 at P_j), which has size 9 (since S_4 contains exactly 9 involutions, all in class $2A$). Thus each vertex has $21 - 9 = 12$ edges leading away from it, and the 12 edges from P distribute over 6 other vertices. By 2-transitivity, exactly $12/6 = 2$ edges connect P to each other vertex Q .

□

3.2 Cheeger Constant

Definition (Random-walk Cheeger constant). For a subset $A \subseteq V$ with $|A| = k \leq \lfloor n/2 \rfloor$, the expansion ratio of A is $h(A) = |\{(s, P) : s \in S, P \in A, s \cdot P \notin A\}| / (|S| \cdot |A|)$. The Cheeger constant is $h = \min$ over all such A of $h(A)$.

Theorem 3.3 (Cheeger constant). *The Cheeger constant of Γ is $h = 8/21$, attained at any cut of size $k = 3$. The discrete Cheeger inequality $h^2/2 \leq \lambda_1 \leq 2h$ holds. Neither bound is an equality.*

Proof. *Computing h .* For a cut of size k , the expansion ratio is $h_k = 2k(7-k)/(|S| \cdot k) = 2(7-k)/21$. Tabulating for $k = 1, 2, 3$:

k	exit pairs	h_k
1	12	12/21
2	20	10/21
3	24	8/21

The minimum is $h = 8/21$, attained at $k = 3$.

Verifying the Cheeger inequality. Lower bound: $h^2/2 = (8/21)^2/2 = 64/882 = 32/441 \approx 0.073 \leq 2/3 = \lambda_1$. ✓

Upper bound: $\lambda_1 = 2/3 = 14/21 \leq 16/21 = 2 \cdot 8/21 = 2h$. ✓

Non-sharpness. Both inequalities are strict: $32/441 < 2/3$ and $14/21 < 16/21$.

□

4. Discussion

4.1 Three Distinct Spectral Quantities

Table 2 summarizes three spectral quantities that arise naturally in the geometry of $G = \text{PSL}(2,7)$ and that must be kept distinct.

Table 2: Spectral quantities

Quantity	Value	Domain	Meaning
λ_1	2/3	$L^2(G/S_4)$, 7-vertex coset graph	Eigenvalue of the coset Laplacian Δ
δ	10/21	$L^2(G)$, full group spectrum	Gap between ρ_6 and ρ_7 eigenvalues
h	8/21	Coset graph $2 \cdot K_7$	Random-walk Cheeger constant

The quantity $\lambda_1 = 2/3$ is a property of the operator Δ on the 7-dimensional coset space, computed in Lemma 2.9. The inter-representation gap $\delta = 10/21$ is the difference between the eigenvalues of Δ on ρ_7 and on ρ_6 in the full group Laplacian Δ^G on $L^2(G)$; it is an algebraic property of the representation theory of $\text{PSL}(2,7)$ that does not directly involve the coset space G/S_4 . The Cheeger constant $h = 8/21$ is an isoperimetric property of the graph Γ , computed in Theorem 3.3. These three quantities are logically independent.

Open Problem 4.1. The Cheeger ratio at 2-vertex cuts is $h_2 = 10/21 = \delta$. No structural explanation for this coincidence is known. In particular, there is no established map between the isoperimetric problem on Γ and the representation-theoretic gap in $L^2(G)$. With the generating set S , the eigenvalues of the full group Laplacian on $L^2(G)$ are 0, 2/3, 1, 8/7 and 4/3 on ρ_1 , ρ_6 , ρ_8 , ρ_7 and ρ_3 , so δ is one of several differences among them (the next eigenvalue above 2/3 belongs to ρ_8); the coincidence should be weighed against that choice.

4.2 The Smith Obstruction

We record a structural constraint on the global topology of orbifolds associated to G , independent of the spectral results above.

Proposition 4.2 (Smith obstruction). *The group $G = PSL(2,7)$ cannot act freely on any sphere S^n for any n . Consequently, S^7/G is an orbifold, not a manifold.*

Proof. The Sylow 2-subgroup of G is isomorphic to D_4 (it appears as the flag stabilizer in Appendix A). This subgroup D_4 contains the Klein four-group $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$, specifically the subgroup generated by any two commuting involutions. The four-group V_4 is a non-cyclic abelian 2-group. By Smith's theorem [11]: if a finite group G contains a non-cyclic abelian p -subgroup, then G cannot act freely on any sphere S^n . Since $V_4 \subset G$ is a non-cyclic abelian 2-subgroup, the conclusion follows. □

Remark 4.3. The Smith obstruction is independent of the spectral gap stability established in Lemma 2.9. The gap stability is a finite-dimensional algebraic fact following from Schur's lemma applied to the coset Laplacian; it requires no topology. The Smith obstruction is a statement about continuous actions on spheres [2]; it does not bear on the eigenvalue $\lambda_1 = 2/3$. These are parallel results about the same group, not causally related.

4.3 Open Problems

Open Problem 4.4 (Orbifold index theory for $S^7/PSL(2,7)$). By Proposition 4.2, the quotient S^7/G , for the action of G on $S^7 \subset \mathbb{R}^8$ through its 8-dimensional irreducible representation ρ_8 , is a 7-dimensional orbifold with non-isolated singular strata: S^3 strata from the 21 class-2A elements (codimension 4) and S^1 strata from classes 3A, 4A, 7A, 7B (codimension 6, total 146 elements; the circle fixed by an element of order 4 lies inside the 3-sphere fixed by its square). Because S^7/G is the quotient of a closed manifold by a finite group, index problems on it reduce to G -equivariant ones on S^7 : the index on G -invariant sections is the average over G of the Lefschetz numbers given by the Atiyah–Singer fixed-point formula, whose fixed-point sets may have any dimension [13]. Kawasaki's index theorem for compact V -manifolds [17], whose complex case is the Riemann–Roch theorem of [8], treats the same problem intrinsically, and groupoid methods [9] extend it further. The η -invariant of $S^7/PSL(2,7)$ is correspondingly the average over G of equivariant η -invariants [14], a finite sum over conjugacy classes computable from the fixed-point data above; it is not computed here.

Open Problem 4.5 (Uniform flag stabilizer description). The three maximal subgroup classes of $PSL(2,7) \cong GL(3,2)$ admit natural incidence-geometric descriptions in $PG(2,2)$. The two S_4 classes are the point-stabilizer $S_4 = \text{Stab}(P_1)$ (Lemma 2.5) and the line-stabilizer $S_4 = \text{Stab}(\ell)$, interchanged by the point-line duality. The Borel subgroup $Z_7 \rtimes Z_3$ is the normalizer of a Singer cycle, fixing no point or line; a constructive $GL(3,2)$ presentation of it is not given in this work. The flag stabilizer $\text{Stab}(P_1, L_{124}) \cong D_4$ (Appendix A) is the Sylow 2-subgroup and is not maximal. A uniform

framework relating these stabilizers within the incidence geometry of $\mathrm{PG}(2,2)$ — and describing how the point-line duality permutes them — would be natural.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting, exposition, and language editing. All mathematical results were verified independently by the author through direct computation. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article. The October 2026 revision (version 3) was prepared with the same assistance; its new counts were checked by explicit computation.

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Appendix A. Constructive $GL(3,2)$ Generators for D_4 as Flag Stabilizer

We realize D_4 as the stabilizer in $GL(3,2)$ of the flag (P_1, L_{124}) , where $P_1 = (1,0,0)$ and $L_{124} = \text{span}_{\mathbb{Z}_2}\{e_1, e_2\} = \{P_1, P_2, P_4\}$ is the projective line through P_1 and P_2 .

Lemma A.1 (D_4 flag stabilizer). *The flag stabilizer $\text{Stab}_{GL(3,2)}(P_1, L_{124})$ is isomorphic to D_4 and is generated over $\mathbb{Z}_2$ by*

$$h_1 = [[1,1,0],[0,1,0],[0,0,1]], \quad h_2 = [[1,0,0],[0,1,1],[0,0,1]]$$

satisfying the presentation $\langle h_1, h_2 \mid h_1^2 = h_2^2 = (h_1 h_2)^4 = I \rangle \cong D_4$.

Proof. We verify each claim by direct computation over $\mathbb{Z}_2$.

(1) *Membership.* $\det(h_1) = 1$ and $\det(h_2) = 1$ over $\mathbb{Z}_2$; both are in $GL(3,2)$.

(2) *Stabilization of P_1 .* $h_1(1,0,0)^T = (1,0,0)^T = P_1$; $h_2(1,0,0)^T = (1,0,0)^T = P_1$.

(3) *Preservation of L_{124} .* The line $L_{124} = \{P_1, P_2, P_4\}$ corresponds to the projective points of $\text{span}(e_1, e_2) = \{0, e_1, e_2, e_1+e_2\}$. The matrix h_1 acts as $(x,y,z) \mapsto (x+y, y, z)$: $h_1(P_1) = P_1$, $h_1(P_2) = P_4$, $h_1(P_4) = P_2$. The matrix h_2 acts as $(x,y,z) \mapsto (x, y+z, z)$: $h_2(P_1) = P_1$, $h_2(P_2) = P_2$, $h_2(P_4) = P_4$. Both generators fix P_1 and map $\{P_1, P_2, P_4\}$ into itself. ✓

(4) *Orders.* $h_1^2(x,y,z) = h_1(x+y, y, z) = (x+y+y, y, z) = (x,y,z)$, so $h_1^2 = I$. $h_2^2(x,y,z) = h_2(x, y+z, z) = (x, (y+z)+z, z) = (x,y,z)$, so $h_2^2 = I$.

(5) *Product order.* $h_1 h_2 = [[1,1,1],[0,1,1],[0,0,1]]$. Let $A = h_1 h_2$. Direct computation gives $A^2 = [[1,0,1],[0,1,0],[0,0,1]] \neq I$ and $A^4 = (A^2)^2 = I$. So $h_1 h_2$ has order 4.

(6) *Non-abelian.* $h_2 h_1 = [[1,1,0],[0,1,1],[0,0,1]] \neq h_1 h_2$.

(7) *Order of the group.* From the presentation $h_1^2 = h_2^2 = (h_1 h_2)^4 = I$ with $h_1 h_2 \neq h_2 h_1$, the group $\langle h_1, h_2 \rangle$ is a non-abelian group of order 8; enumerating $\{I, h_1, h_2, h_1 h_2, (h_1 h_2)^2, h_1(h_1 h_2)^2, (h_1 h_2)^3, h_1(h_1 h_2)^3\}$ shows $|\langle h_1, h_2 \rangle| = 8$. Since Q_8 has a unique element of order 2 but h_1 and h_2 are two distinct involutions, $\langle h_1, h_2 \rangle \cong D_4$. ✓

The index $|G|/|\langle h_1, h_2 \rangle| = 168/8 = 21$ equals the number of flags (incident point-line pairs) in $PG(2,2)$, confirming that $\langle h_1, h_2 \rangle$ is the full flag stabilizer. It is a Sylow 2-subgroup of G , contained in S_4 and therefore not maximal (cf. Remark 2.2).

□